

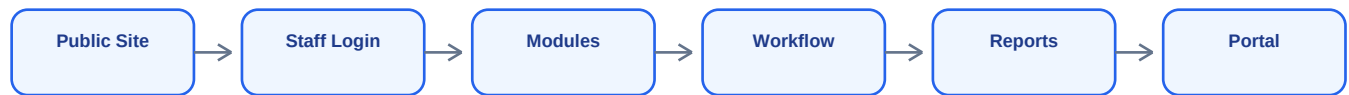
Hamro Hospital Management System

Complete Project Documentation

1. Project Overview

Hamro Hospital Management System is a Django-based integrated HIS covering patient registration, OPD, doctor consultations, referrals, billing, finance, laboratory, radiology, pharmacy, admissions, nursing, operation theatre, blood bank, insurance, medical records, staff attendance and payroll. The UI design is preserved while workflows are integrated.

2. System Architecture



Module	Summary
Accounts & Staff	Users, roles, staff ID/barcode, attendance, leave, payroll and staff benefits.
Patients & OPD	Registration, QR/barcode, OPD visits, tickets, patient history sections.
Doctors	Doctor profiles, dashboard, patient search, consultations and referrals.
Workflow	ServiceOrder, PaymentEvent, PatientTimeline and department order queues.
Billing & Finance	Cash Counter, pending payments, receipts, finance reports and payroll access.
Lab/Radiology	Requests, verification, result uploads, templates and structured lab values.
Pharmacy	Prescription queue, sales, stock validation, batches and ledger.
Admission/Nursing/OT	Admissions, beds, deposits, transfers, discharge checklist/package, nursing notes, surgery workflow.
Blood Bank/Insurance	Blood requests/issues, compatibility, inventory lifecycle, claims lifecycle.
Website/Portal	Public website, extension services, patient portal, online bill/visit flows.

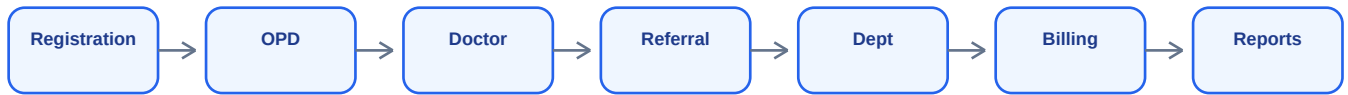
3. Installation Guide

Install Python 3.11/3.12, create a virtual environment, install requirements, run migrations, create demo accounts and start the server. Commands: `pip install -r requirements.txt`, `python manage.py migrate`, `python manage.py create_demo_accounts`, `python manage.py runserver`.

4. User Roles and Permissions

Role	Key access
Super Admin	Full system, settings, users, reports, all records.
Finance/Accounts	Billing, revenue, attendance, payroll, salary, payment records.
Department Head	Own department staff, attendance, leave, referrals, service orders and reports.
Registration	Patient registration, cards, OPD and patient details.
Cashier	Pending payments, receipts and collections.
Doctor	Clinical patient view, consultations, referrals and own profile.
Departments	Own operational queues and records.
Patient	Own portal records, bills, visits and documents.

5. Major Workflows



Barcode-driven patient identification is used across registration, OPD, billing, departments and printed documents. Referrals create ServiceOrders and pending bills; payments create PaymentEvents and update departments.

6. Database Overview

Core entities include User, Patient, Visit, Doctor, Referral, ServiceOrder, Bill, PaymentEvent, PatientTimeline, Admission, Ward, Bed, Medicine, LabTestRequest, RadiologyRequest, InsuranceClaim, StaffAttendance, StaffLeaveRequest and StaffSalaryPayment.

7. Backup, Restore and Maintenance

Use `backup_data` and `restore_data` commands, PostgreSQL dumps for production, media backups, offsite storage, routine restore tests and versioned deployment.

8. Troubleshooting

Run `python manage.py check`, `python manage.py migrate`, `python manage.py verify_production_config`, inspect server logs, confirm environment variables and verify Redis/Celery/eSewa settings.